Theorem 1.9

If A is a non-singular square matrix of order n, then

(i) $(\text{adj } A)^{-1} = \text{adj}(A^{-1}) = \frac{1}{|A|}A$ (ii) $|\text{adj } A| = |A|^{n-1}$

(iii) $\text{adj}(\text{adj } A) = |A|^{n-2} A$ (iv) $\text{adj}(\lambda A) = \lambda^{n-1}\text{adj}(A)$, λ is a nonzero scalar

(v) $|\text{adj}(\text{adj}A)| = |A|^{(n-1)^2}$ (vi) $(\text{adj } A)^T = \text{adj}(A^T)$

Proof

Since A is a non-singular square matrix, we have $|A| \neq 0$ and so, we get

(i) $A^{-1} = \frac{1}{|A|}(\text{adj } A) \Rightarrow \text{adj } A = |A| A^{-1} \Rightarrow (\text{adj } A)^{-1} = \left(|A| A^{-1}\right)^{-1} = \frac{1}{|A|}\left(A^{-1}\right)^{-1} = \frac{1}{|A|}A.$

Replacing A by A^{-1} in $\text{adj } A = |A| A^{-1}$, we get $\text{adj}(A^{-1}) = |A^{-1}|\left(A^{-1}\right)^{-1} = \frac{1}{|A|}A.$

Hence, we get $(\text{adj } A)^{-1} = \text{adj}(A^{-1}) = \frac{1}{|A|}A.$

(ii) $A(\text{adj } A) = (\text{adj } A)A = |A| I_n \Rightarrow \det(A(\text{adj } A)) = \det((\text{adj } A)A) = \det(|A| I_n)$

$\Rightarrow |A||\text{adj } A| = |A|^n \Rightarrow |\text{adj } A| = |A|^{n-1}.$

(iii) For any non-singular matrix B of order n, we have $B(\text{adj } B) = (\text{adj } B)B = |B| I_n.$

Put $B = \text{adj } A$. Then, we get $(\text{adj } A)(\text{adj}(\text{adj } A)) = |\text{adj } A| I_n.$

So, since $|\text{adj } A| = |A|^{n-1}$, we get $(\text{adj } A)(\text{adj}(\text{adj } A)) = |A|^{n-1} I_n.$

Pre-multiplying both sides by A, we get $A\left((\text{adj } A)(\text{adj } (\text{adj } A))\right) = A(|A|^{n-1} I_n).$

Using the associative property of matrix multiplication, we get

$(A(\text{adj } A))\text{adj}(\text{adj } A) = A(|A|^{n-1} I_n).$

Hence, we get $(|A| I_n)(\text{adj}(\text{adj } A)) = |A|^{n-1} A.$ That is, $\text{adj}(\text{adj } A) = |A|^{n-2} A.$

(iv) Replacing A by λA in $\text{adj}(A) = |A| A^{-1}$ where λ is a non-zero scalar, we get

$$\text{adj}(\lambda A) = |\lambda A|(\lambda A)^{-1} = \lambda^n |A| \frac{1}{\lambda} A^{-1} = \lambda^{n-1} |A| A^{-1} = \lambda^{n-1}\text{adj}(A).$$

(v) By (iii), we have $\text{adj}(\text{adj } A) = |A|^{n-2} A$. So, by taking determinant on both sides, we get

$$|\text{adj}(\text{adj}A)| = \left||A|^{n-2} A\right| = \left(|A|^{(n-2)}\right)^n |A| = |A|^{n^2-2n+1} = |A|^{(n-1)^2}.$$

(vi) Replacing A by A^T in $A^{-1} = \frac{1}{|A|}\text{adj } A$, we get $\left(A^T\right)^{-1} = \frac{1}{|A^T|}\text{adj}(A^T)$ and hence, we get

$$\text{adj}(A^T) = |A^T|\left(A^T\right)^{-1} = |A|\left(A^{-1}\right)^T = \left(|A| A^{-1}\right)^T = \left(|A| \frac{1}{|A|}\text{adj } A\right)^T = (\text{adj } A)^T.$$

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